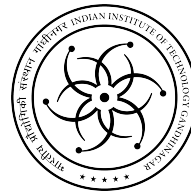


Meduri Ruthwick

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Education

- 2025 - Ongoing **Indian Institute of Technology Gandhinagar,**
Biological Engineering, Ph.D.
GPA – 9.01/10
- 2023 - 25 **Indian Institute of Technology Gandhinagar,**
Biological Engineering, Master of Technology.
GPA – 9.79/10
- 2017 - 2021 **Chaitanya Bharathi Institute of Technology,**
Biotechnology, Bachelor of Technology.
GPA – 8.65/10

Academic Projects

- Jun 2025 – Ongoing **Deep analysis of FANTOM CAGE data reveals hierarchical patterns of TSS co-deployment hubs and their disruption in cancers,**
Supervisor: Dr. Umashankar Singh, Indian Institute of Technology Gandhinagar.
- Autoencoder based analysis of CAGE data reveals that normal cells employ epromoter-like core TSSs to coordinate multi-promoter transcription efficiently, whereas cancers disrupt this parsimony with widespread distal TSS activation.
- Jan 2024 – May 2025 **Predictive application of ATAC-seq data in evolution of genome regulation,**
Supervisor: Dr. Umashankar Singh, Indian Institute of Technology Gandhinagar.
- Studying the evolution, origins, and biology of open chromatin regions in primate genomes using a machine learning model based on sequence features from ATAC-seq data, with potential extension to vertebrate genomes as well.
- Mar 2021 – May 2021 **Memory Consolidation and its Relationship with Dream Content,**
Supervisor: Dr. Supratim Ray, Indian Institute of Science Bangalore.
- Explored memory consolidation mechanisms during REM and non-REM sleep to enhance our understanding of memory processing

Independent Projects

- Aug 2023 – Ongoing **Biology and Evolution of CRISPR systems in Prokaryotes,**
Key Words: CRISPR Biology, Spacer evolution, Sequence entropy.
- Investigating how sequence entropy variations and spacer dynamics within CRISPR regions contribute to prokaryotic adaptation and defense.
 - Exploring the impact of entropy on spacer diversity, insertion site preferences, and overall CRISPR system evolution in microbial genomes.

Achievements

- Jun 2025 **Institute gold medal for highest performer in M.Tech class of 2025 in Biological Sciences and Engineering, IIT Gandhinagar.**
- Mar 2020 **Selected for prestigious fellowships from JNCASR & INASc.**
- Worked on a proposal titled "Memory Consolidation and its Relationship with dream content"
- Mar 2020 **Secured a grant of Rs. 1,00,000 for a proposal developed as a part of BIRAC-BIIS 6 organised by DBT, Government of India.**
- Worked on a proposal titled "Efficient way of development of seedlings using sand"

Feb 2020 **Secured Best Poster Award (3rd) in conference "*Biotechnology and Bioengineering Trends*" organised by JNTU Hyderabad.**

- Presented a poster titled "*In-silico study of Cognitive Behaviour in relation with Implications of Differential Gene Expression controlled by Brain*"

Jun 2017 - Aug 2021 **Secured several best paper awards during inter collegiate level symposium.**

- Presented papers on topics ranging *Genetic Basis of Fingerprints, In-silico study of Cognitive Behaviour in relation with implications of differential gene expression controlled by brain, Human Brain A Mystery Unsolved, Industrial Therapeutic Cloning*

Skills

Microbiology	Microscopy, Bacterial Culture techniques, Gram staining, Microbial Identification, Microbial Physiology assays.
Genetic Engineering	DNA extraction, Polymerase Chain Reaction (PCR), Gel Electrophoresis, Cloning, Transformation
Cell Culture	Cell seeding, Passaging, Cell counting, Viability Assays, Cryopreservation, Sterile technique, Media preparation, Cell Culture maintenance, Transfection, Infection, Microscopy [Under Training].
Computational Biology	Autodock Vina, Galaxy Server [Genome Analysis, Transcriptome Analysis, Peak Calling using MACS2 & HMMRATAC], Primer Design.
Protein Biology	Protein Biology Laboratory Skills: SDS PAGE, MALDI TOF, LC/GC - MS.
Statistical Analysis & Coding	Python, C, R

Relevant Courses

Classroom	Molecular Oncology, Biochemistry(Advanced), Molecular and Cellular Biotechnology, Biostatistics, Microbiology, Immunology, Immuno Diagnostics
Online	Immunology - Host Pathogen Interactions [NPTEL], Epidemics - The Dynamics of Infectious Diseases, Computational Genomics [NPTEL], Science of Stem Cells, Advanced Neurobiology I & II

Positions of Responsibilities & Extra Curricular Activities

Aug 2023 - Ongoing	Teaching Associate , <i>Department of Biological Sciences and Engineering, IITGn.</i>
	Course Assistant - Biology for Engineers (Jun'24 - Present), Departmental Activities (Jan '24 - May '24), Foundation Program (Aug '23 - Dec '23)
Sept 2023 - Ongoing	Academic Representative , <i>Department of Biological Sciences and Engineering, IITGn.</i>
	Representing M.Tech 2023 cohort to the department regarding academic-related constraints
Sept 2023 - Aug 2024	Institute Senate - IIT Gandhinagar , <i>Student Member.</i>
	Representing M.Tech cohort of IIT Gandhinagar regarding academic and otherwise related constraints